FES Code4Nature Challenge 2026

Ideation / Concept Note — EcoWatch

TEAM INFORMATION

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Theme: Local Governance

Problem Statement

The Core Problem

Environmental issues such as **garbage accumulation, illegal dumping, dust, construction debris, and water pollution** are common in public spaces. However, these issues are often reported through informal channels such as word of mouth, phone calls, messages, or manual complaints. As a result, there is often **no centralized system** to record what was reported, where it occurred, who was assigned to handle it, whether the issue was resolved, and how long the process took.

This creates several interconnected challenges:

1. Fragmented Monitoring and Reporting

- Environmental complaints are scattered across different channels, making it difficult to maintain a **single, up-to-date record** of reported issues.
- There is no clear and consistent timeline showing whether an issue is **reported, assigned, in progress, cleaned, verified, or closed**.

2. Limited Accountability and Verification

- Once an issue is assigned for cleanup, it can be difficult to track **who is responsible, whether the task was completed on time, and whether the reported problem was actually resolved**.
- The absence of reliable **before-and-after evidence** makes it difficult for supervisors to verify completed cleanup work.

3. Recurring Issues and Inefficient Resource Planning

- The same locations may experience repeated garbage dumping or other environmental problems, but without **historical and location-based data**, recurring hotspots and possible causes are difficult to identify.
- Limited information about issue frequency, response time, and cleanup activities makes it harder to **plan workers, equipment, and resources effectively**.

Therefore, there is a need for a **centralized, transparent, and data-driven system** that connects environmental reporting with task assignment, cleanup tracking, evidence-based verification, and historical monitoring. **EcoWatch is designed to address this gap by providing an end-to-end workflow from reporting an environmental issue to its verified resolution and long-term monitoring.**

Solution Overview

The Solution, in One Line

EcoWatch transforms scattered environmental complaints into a centralized, location-based workflow for reporting, assignment, cleanup, verification, and long-term monitoring.

Environmental issues are often reported through different channels, making it difficult to track an issue from the initial complaint to its final resolution. **EcoWatch brings reporting, location mapping, task assignment, field operations, evidence collection, and supervisor verification into one integrated platform.**

The system provides a clear workflow for communities, field workers, supervisors, and other responsible teams, ensuring that every reported issue can be **tracked from submission to verified closure**.

Core Workflow

• **Report —** Citizens or authorized field staff submit an environmental issue with its category, description, location, severity, and supporting photograph.

• **Assign & Track —** The reported issue appears on a centralized dashboard, where a supervisor can review it and assign the task to an appropriate field worker. The system tracks the task status, assigned worker, and deadline.

• **Clean & Verify —** The field worker completes the required action and uploads before-and-after photographs as evidence. The supervisor reviews the evidence and verifies whether the issue has been successfully resolved before closing the task.

Here is what each part of the platform does:

Platform Part	What it does
Issue Reporting	Allows users to submit environmental issues with category, description, location, severity, and supporting photographs.
Live Map Dashboard	Displays reported issues geographically according to their location, severity, and current status.
Task Assignment	Enables supervisors to assign reported issues to appropriate and available field workers.
Worker View	Provides field workers with assigned tasks, locations, deadlines, status updates, and evidence-upload functionality.
Before/After Proof	Captures visual evidence of the reported location before and after the cleanup activity for verification.
Status & Accountability Tracking	Maintains the history of reporting, assignment, progress, completion, verification, and closure.
Historical Monitoring	Stores location-based issue history to identify recurring problems and support better planning and preventive action.

Challenge-fit concept: EcoWatch is designed as a **web-based MVP** that can initially demonstrate the complete reporting-to-resolution workflow using representative or sample data. As the system develops, it can be extended with l**ive location services, cloud-based image storage, AI-assisted verification, automated notifications, real-world environmental data, and advanced analytics.**

The core objective is to move environmental issue management from **“report and forget”** to a **trackable, verifiable, and data-driven process.**

Innovation Highlights

The key innovation of EcoWatch is that it goes beyond basic environmental issue reporting by creating a **complete, traceable, and data-driven lifecycle** from reporting to verified resolution. The platform combines community reporting, field-level action, evidence-based verification, and long-term environmental monitoring.

1. AI-Assisted Before–After Verification

EcoWatch uses AI-assisted image comparison to analyze the **before-and-after photographs** submitted by field workers and generate a cleanup confidence score. This helps supervisors identify whether a significant change is visible after the cleanup, while the **final verification remains with the supervisor**.

2. Duplicate Report Detection

When multiple users report the same environmental issue, EcoWatch can identify **possible duplicate reports** by comparing factors such as location, issue category, and reporting time. Instead of creating unnecessary duplicate tasks, the system flags the reports for the supervisor to review and either merge or keep separate.

3. Automatic Escalation

EcoWatch assigns a **deadline** to cleanup tasks based on their priority or severity. If a task is not accepted or completed within the required time, the system can automatically flag or escalate it to the supervisor, helping reduce delays and improve accountability.

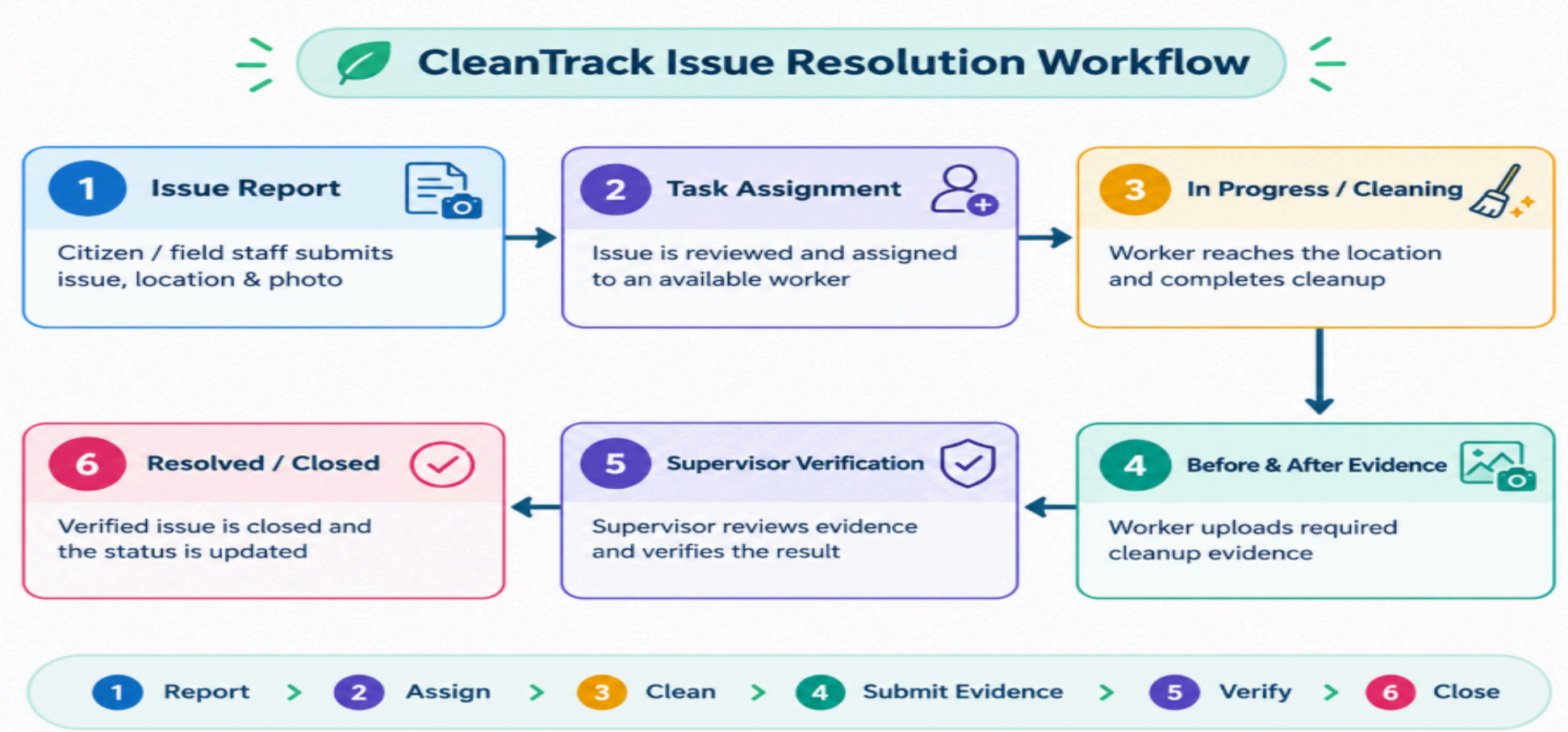
4. Recurrence Prevention Tracker

EcoWatch does not only track whether an issue was cleaned; it also monitors whether **the same problem returns at the same location**. Repeated incidents can be linked with possible causes, such as insufficient waste bins or irregular collection, and preventive actions can be recorded and monitored to determine whether recurrence decreases.

5. Waste Destination Tracking

After completing a cleanup, workers can record the **type and estimated quantity of waste collected and its destination**, such as recycling, composting, reuse, or landfill. This provides additional insight into how collected waste is being managed and gives EcoWatch a stronger focus on sustainable waste management.

reported, who handled it, what action was taken, and whether the cleanup was verified. The collected historical and location-based data can also help identify recurring environmental hotspots and support better planning of workers and resources.



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Core Workflow

Report → Duplicate Check → Assign → Clean → Submit Evidence → AI-Assisted Check → Verify → Waste Tracking → Close → Monitor Recurrence

Prototype Differentiator

Rather than stopping at **“issue reported”** or **“issue resolved,”** EcoWatch creates a complete and verifiable lifecycle. It records **what was reported, where it occurred, who handled it, when it was handled, what evidence was submitted, how the result was verified, and whether the problem returns later**.

Impact Assessment

EcoWatch is designed to improve environmental issue management by making the process more transparent, accountable, verifiable, and data-driven. Different stakeholders benefit at different stages of the reporting, response, cleanup, and monitoring process.

Primary Stakeholder	How EcoWatch helps

Community Residents	Report environmental issues, track their progress without exposing unnecessary personal information, and see verified outcomes.
Field Workers	Receive clearly assigned tasks with locations and deadlines, complete cleanup activities, and submit before-and-after evidence.
Supervisors / Local Agencies	Monitor reported issues, assign and track tasks, receive escalation alerts for delayed work, verify cleanup evidence, and evaluate response performance.
NGOs / Community Groups	Use mapped reports, historical records, and recurring-issue data to plan cleanup drives, awareness campaigns, and community interventions.
Decision-Makers	Use environmental trends, recurring hotspots, response times, and waste-management data to support better resource allocation and targeted preventive action.

Expected Impact

- **Improved accountability:** Every issue can be tracked from reporting to verified closure.
- **Faster response:** Task assignment, deadlines, and automatic escalation can reduce delays in addressing environmental issues.
- **Reliable verification:** Before-and-after evidence, supported by AI-assisted analysis, provides stronger evidence of cleanup activities.
- **Prevention of recurring issues:** Historical location-based data can help identify repeated problems and support preventive measures rather than repeated cleanup alone.
- **Better resource planning:** Issue trends, severity, response time, and hotspot data can help organizations allocate workers, equipment, and resources more effectively.

- **Improved waste management:** Recording waste type, quantity, and destination provides insight into how collected waste is being handled.
- **Greater transparency and trust:** A clear record of actions and verified outcomes can improve confidence among communities and responsible organizations.

Data & Hardware Requirements

Potential Data Inputs

- Reported pollution/waste issues including category, severity, location, status, and timestamps
- GPS coordinates for reported issues and worker check-ins
- Before-and-after cleanup images with timestamps and location information
- Task assignments, status history, cleanup duration, and verification records
- Historical issue data for hotspot and trend analysis

Prototype Hardware

- A standard laptop or desktop with internet access
- A smartphone/mobile device with GPS and camera for field workers
- Internet/mobile data connectivity for reporting and photo uploads

Prototype Software Stack

- React / JavaScript for the web dashboard
- Node.js / Express for backend APIs
- MongoDB or another suitable database for reports and task records
- Leaflet / Google Maps API for location and map visualization
- Cloud image storage such as Firebase Storage, Cloudinary, or AWS S3

Scalability Note

Real-world deployment can progressively connect stronger location services, cloud storage, notifications, analytics, and additional verification mechanisms as data availability and operational requirements grow.